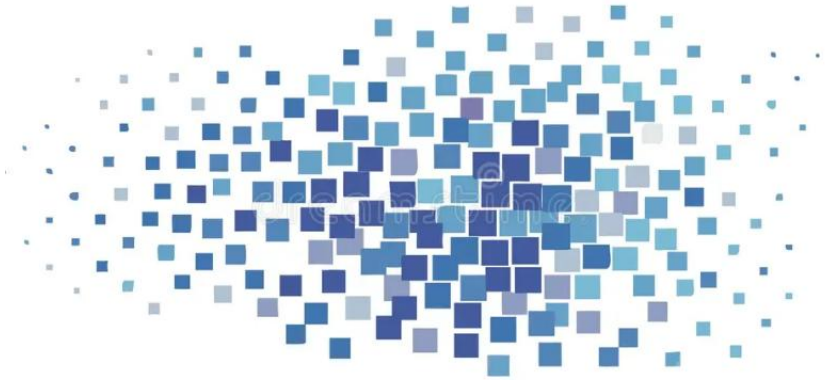
Information Diffusion in Social Network

(SNA – B Tech III CSE)

(Week#8 – Feb 25, 2026)

Dr Dhiren Patel



Diffusion

- Diffusion is simply how something spreads through a network—disease, rumor, or Information.
- Every day, diffusion is happening around you.
- Instagram reels spread.
- Memes spread.
- News spreads.
- Even job opportunities spread.
- Why do some things spread to millions… while others die immediately?

Diffusion

- Suppose diffusion starts from central hub.
- How many people get it immediately? Many.
- And if diffusion starts from edge node? Few..
- Same information. Same network. Only starting position changed.
- Diffusion is controlled by structure.
- This is why companies pay influencers, because they occupy powerful positions in the network..
- Same idea explains disease spread.

Questions - Goals

- First: How do we identify the most powerful nodes in a network?
- Second: How do we use them to maximize diffusion?
- Third: How do we stop diffusion when it is dangerous?

Measuring diffusion

- **Metrics**
- **infection size** - (Final Reach) **-** Total number (or fraction) of nodes that become infected (activated) during the diffusion process. (e.g. Viral marketing success)
- **cascade depth** – (Maximum Spread Distance) **-** Maximum number of steps from initial seed node to farthest infected node.
- **reproduction number** - Average number of new infections caused by each infected node. (Measures spread intensity).
- **Modularity** – (Community Containment Measure) - Measures how strongly diffusion is contained within communities.
- **Conductance** – (Community Boundary Strength) - Measures how easily diffusion escapes a community.

Metrics – cont.

- **Centrality** - (Influence Measure) **-** Measures importance of nodes in diffusion.
- **Degree centrality** - Measures immediate spread potential.
- **Betweenness centrality** - Measures control over diffusion paths.
- **Closeness centrality** - Measures speed of reaching network.

Summary table

Metric	**Definition**	**Measures**
Infection size	total infected nodes	reach
Cascade depth	max distance from seed	penetration
Reproduction number	new infections per node	spread intensity
Modularity	community containment	echo chambers
Conductance	boundary strength	escape difficulty
Centrality	node importance	influence

Example – disease

Metric	**Meaning**
Infection size	total infected people
Cascade depth	longest infection chain
R_0 - Reproduction number	infections per person
Modularity	infection trapped in city
Conductance	travel between cities
Centrality	super-spreaders

If one infected person infects 10 others vs 1 other, which spreads faster?

Intervention strategies

- **Question**
- “How do we control diffusion?”

Goal	**Strategy**
speed spread	seed hubs
cross groups	seed bridges
stop rumor	cut bridges
break echo chamber	add weak ties

Weak ties connect communities and allow new information to escape local bubbles.

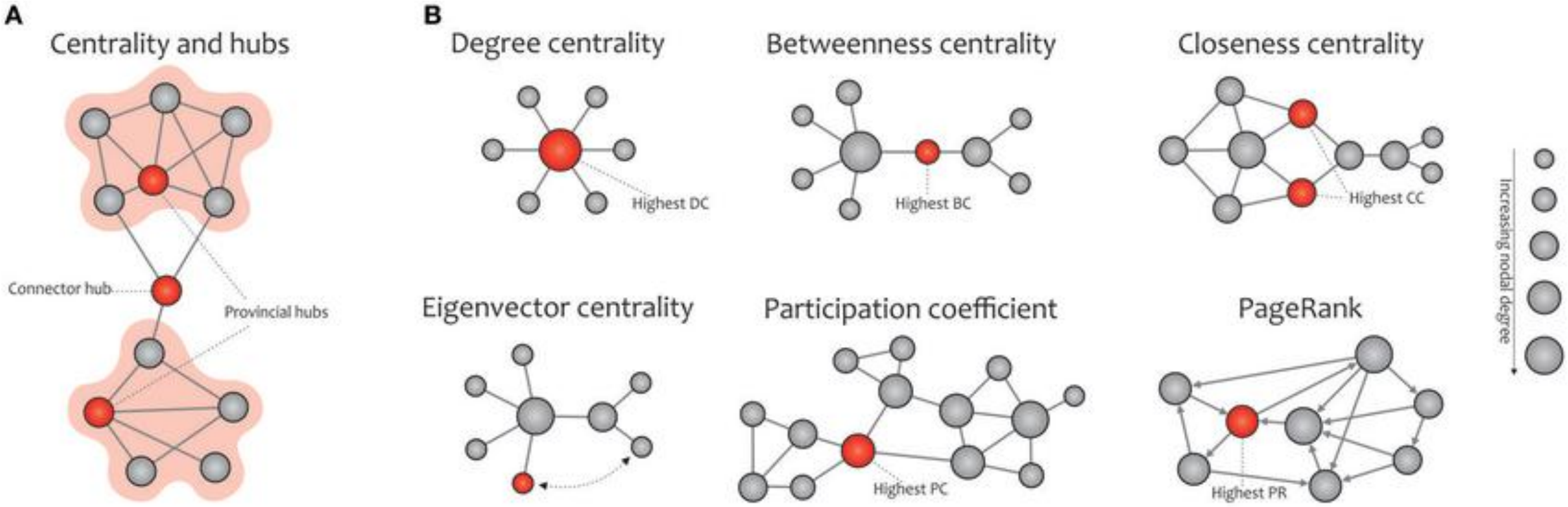
Cont.

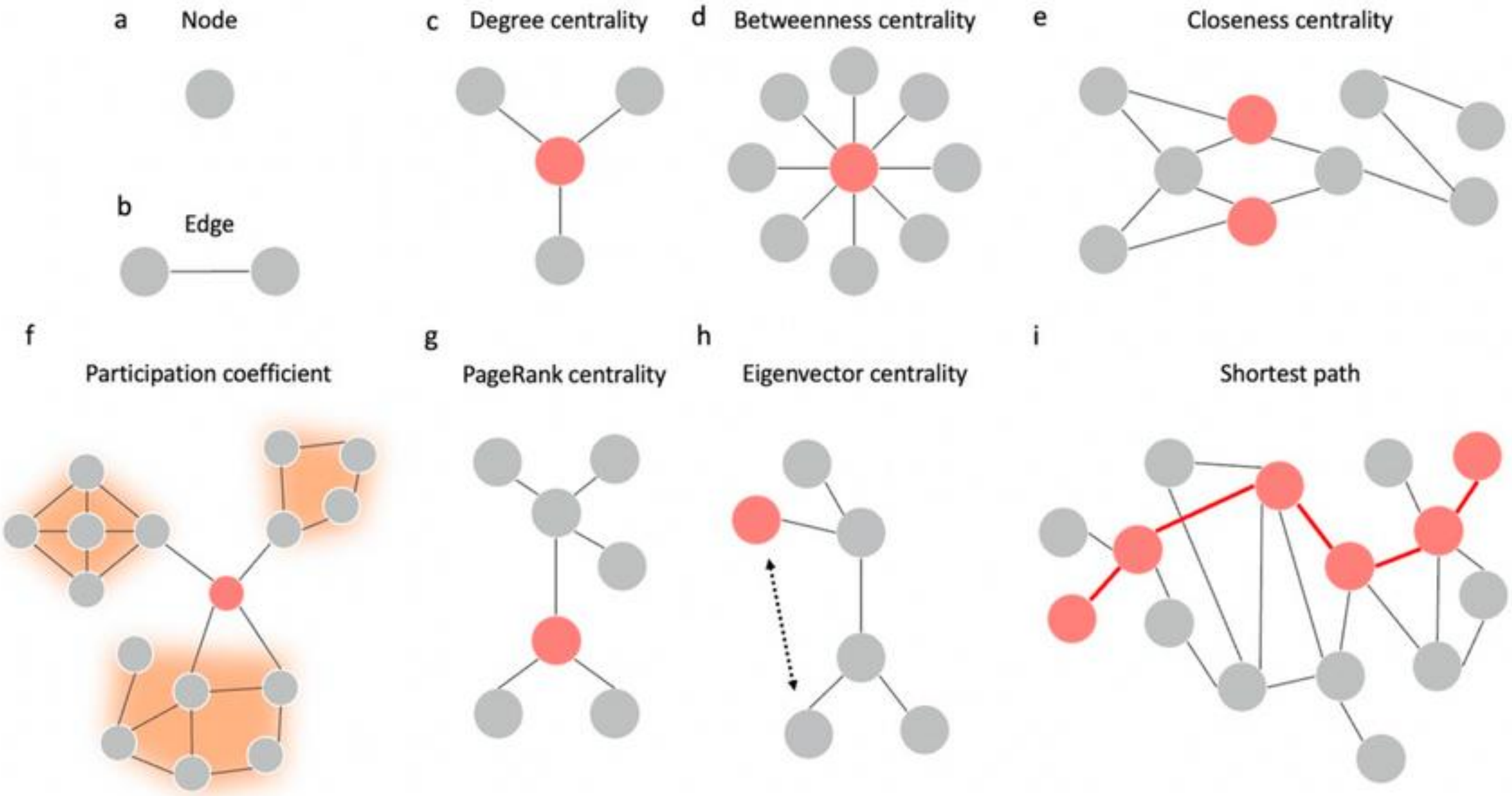
- **Seed hubs** - Start diffusion from high-degree nodes to maximize immediate reach (Metric: Degree centrality, Use case – Viral marketing)
- **Seed bridges** - Start diffusion at nodes that connect communities to cross structural boundaries (Metric: Betweenness centrality, Use case - breaking echo chambers)
- **Immunize hubs** - Protect or remove high-degree nodes to stop rapid large-scale spread (Metric: Degree centrality, Use case - epidemic control)
- **Cut bridges** - Remove or monitor bridge edges to isolate communities and contain diffusion locally (Metric: Betweenness (nodes or edges), Use case - quarantine strategies)

Structure Controls Spread

- Degree centrality → spread speed
- Betweenness centrality → spread reach
- Closeness centrality → spread efficiency

- Diffusion is controlled by where you start, not just what you spread.
- Hub → many edges → fast spread


A
Centrality and hubs
Connector hub
Provincial hubs
B
Degree centrality
Highest DC
Betweenness centrality
Highest BC
Closeness centrality
Highest CC
Eigenvector centrality
Participation coefficient
Highest PC
PageRank
Highest PR
Increasing nodal degree


a
Node
b
Edge
c
Degree centrality
d
Betweenness centrality
e
Closeness centrality
f
Participation coefficient
g
PageRank centrality
h
Eigenvector centrality
i
Shortest path

Method

- **Seed hubs strategy** means selecting nodes with the highest degree centrality as initial diffusion sources to maximize immediate reach in the network.
- Large central nodes = hubs
- Many edges emanating outward
- Diffusion spreads fastest from hubs
- Peripheral nodes reach fewer neighbors

Mathematical Diffusion Gain

Expected spread proportional to degree:

$$Spread(v) \propto deg(v)$$

For seed set S:

$$Total\ Reach \approx \sum_{v \in S} deg(v)$$

Maximized when selecting highest-degree nodes.

Use Case: Viral Marketing

- Goal: Promote product
- Bad strategy: Seed random users → slow spread
- Good strategy: Seed influencers (high-degree nodes)
- Result: Fast viral spread
- E.g.
- Instagram influencers
  Twitter users with many followers
  YouTube creators with many subscribers

Seed Hubs Strategy
Metric: Degree centrality
Definition:
Select nodes with highest number of connections as diffusion seeds.
Why:
Maximizes immediate reach.

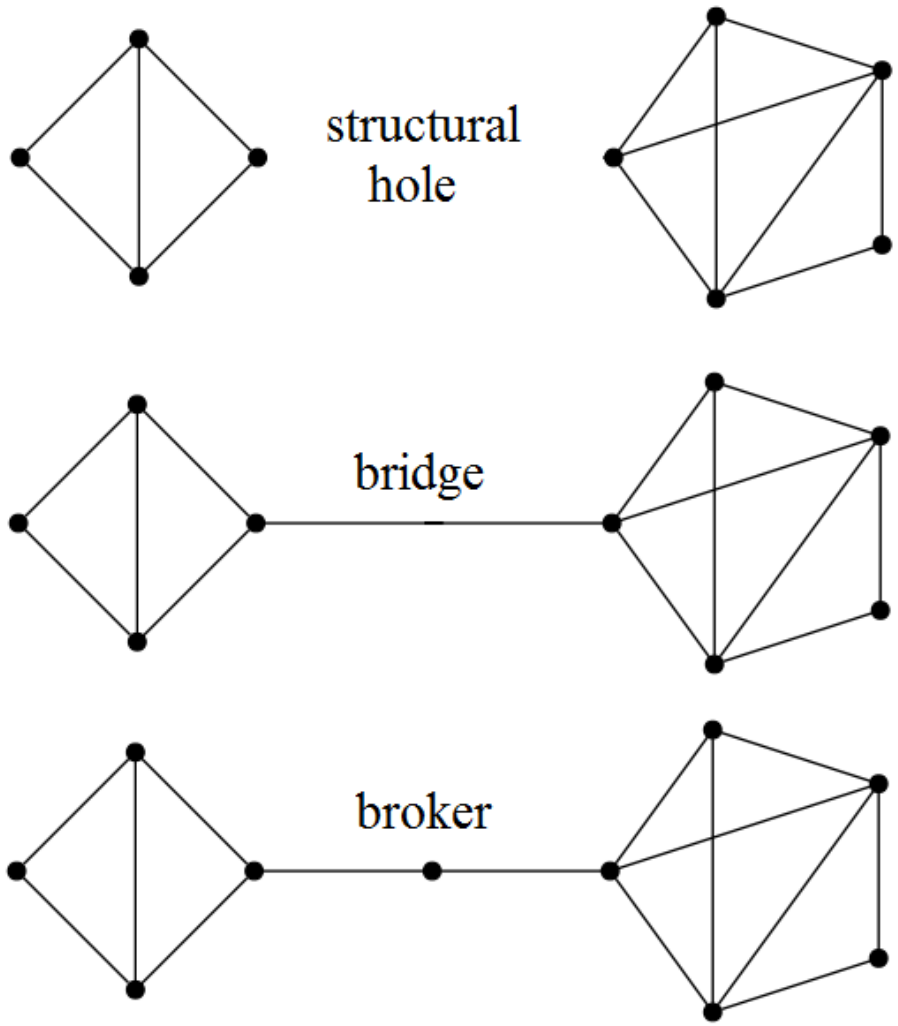
How?

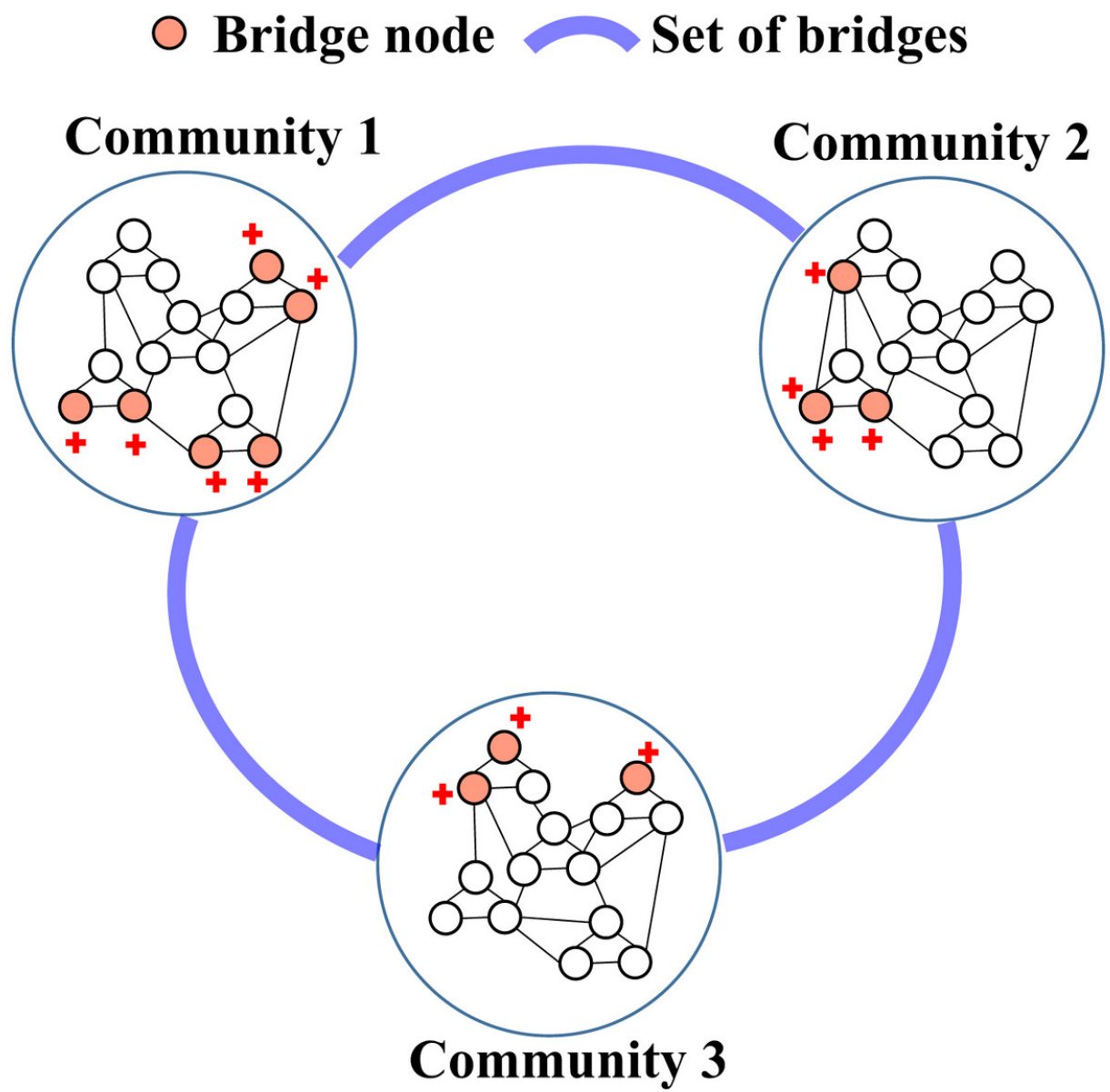
- **Algorithm**
- Step 1: Compute degree for all nodes
  Step 2: Sort nodes descending by degree
  Step 3: Select top k nodes

- Compute degree → sort → pick top k

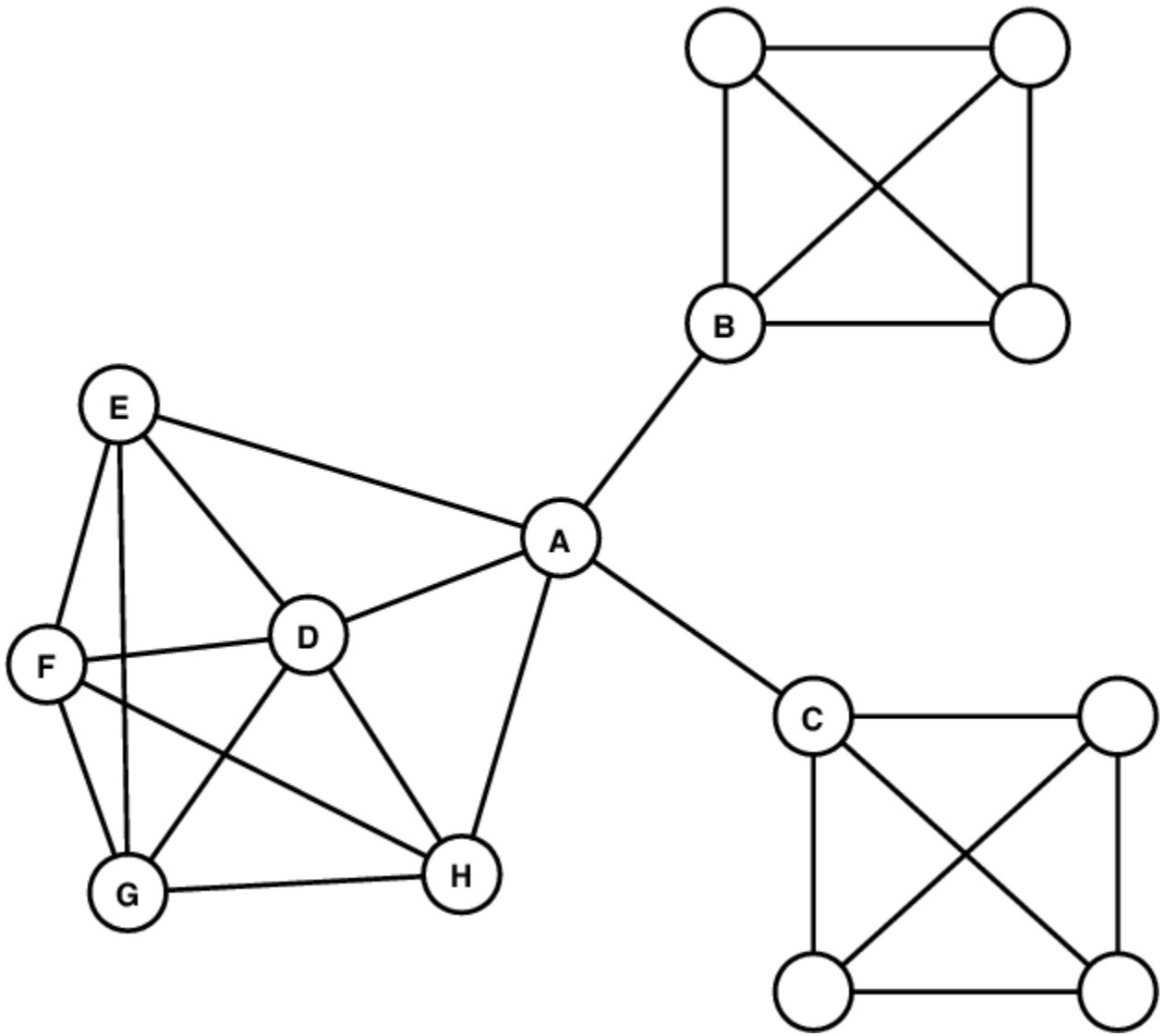
Method

- Hubs spread fast locally. Bridges spread far globally.
- Bridge connects communities → diffusion crosses boundaries.
- **Seed bridges strategy** selects nodes that lie on many shortest paths between communities, enabling diffusion to cross structural boundaries.
- Metric used: **Betweenness centrality**
- **Algorithm**
- Step 1: Compute betweenness centrality for all nodes
  Step 2: Select nodes with highest betweenness
  Step 3: Use as seed nodes


structural
hole
bridge
broker


Bridge node
Set of bridges
Community 1
Community 2
Community 3

- node A's bridging role



Seed bridge

- Case 1: Seed hub inside community C1
- Result:
- Community C1 → fully infected
  Community C2 → not reached
- Case 2: Seed bridge node
- Result:
- Community C1 → reached
  Community C2 → reached
- Bridge enables global diffusion.

Seed bridge

- **Why bridges are powerful**
- If you seed inside one community:
- Diffusion stays trapped locally.
- If you seed a bridge:
- Diffusion jumps to new community.
- This breaks structural barriers.
- Bridges connect otherwise disconnected groups.

Use Case: Breaking Echo Chambers

- Echo chamber structure:
- Group A ←→ internal connections
- Group B ←→ internal connections
- Seeding bridge nodes: Information crosses echo chambers.
- Used in:
- Public awareness campaigns
  Political messaging
  Counter misinformation

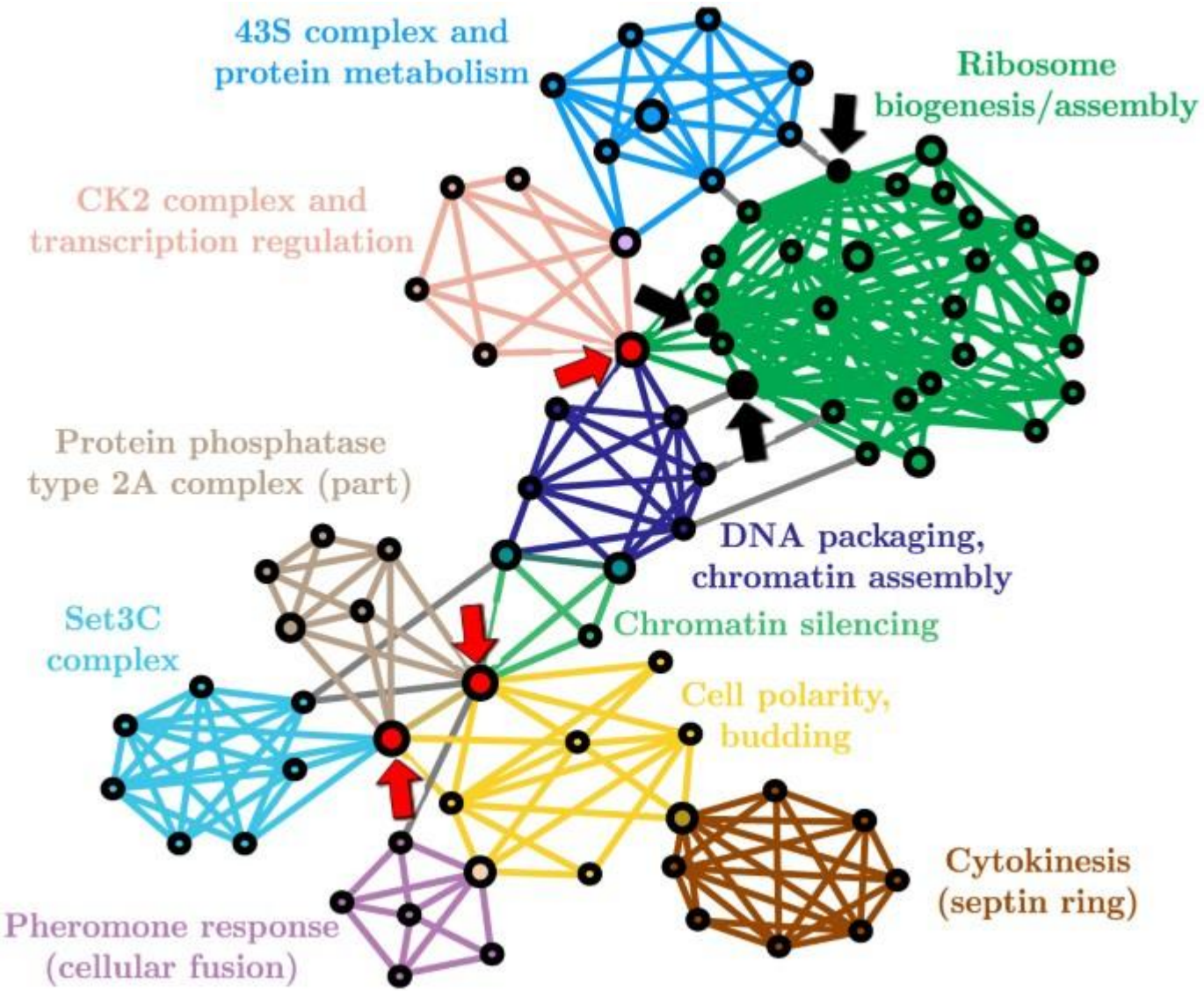
Real-world Examples

- Twitter users followed by multiple communities
- Journalists connecting political groups
- Researchers collaborating across disciplines
- Employees connecting departments
- **Seed Bridges Strategy**
- Metric: Betweenness centrality
- Definition: Select nodes that connect communities.
- Why: Enables diffusion across structural boundaries.

Summary table

Strategy	**Metric**	**Goal**
Seed hubs	Degree centrality	Maximize immediate local reach
Seed bridges	Betweenness centrality	Maximize global cross-community reach

diffusion cannot reach other clusters unless it passes through bridge nodes



- Communities (clusters) → colored groups
- Dense internal connections → many edges within each group
- Few inter-community connections → limited edges between groups
- Red nodes are High betweenness centrality nodes (They act as connectors between otherwise separate clusters)
- if bridge nodes are removed - network fragmentation happens
- **Hubs give speed. Bridges give reach.**

Method for Stopping Diffusion

- **Immunize hubs strategy** protects or removes nodes with highest degree centrality to prevent rapid large-scale diffusion.
- Metric used: **Degree centrality**
- **Why hubs are critical in epidemic spread**
- Hubs have many connections.
- If infected, they infect many nodes quickly.
- If H infected → infects entire network rapidly.
- After immunization:
- Hub removed or protected: Network fragments: And Spread stops.

Targeted vs Random Immunization

- Random immunization:
- Select random nodes → inefficient
- Targeted immunization:
- Select highest-degree nodes → extremely effective
- E.g. Vaccinate healthcare workers first
- Before hub immunization: Infection → Hub → Entire network
- After hub immunization: Infection → stops locally
- Network-wide epidemic prevented.

How?

- **Algorithm**
- Step 1: Compute degree centrality for all nodes
  Step 2: Sort descending
  Step 3: Immunize top k nodes
- **Use Case: Epidemic Control**
- Goal: stop disease spread
- Strategy: Vaccinate high-degree individuals
- Result: Rapid epidemic containment

Real-world Example: COVID Vaccination

- Vaccinating:
- Healthcare workers, Airport staff, Public transport workers
- These are hubs in contact network.
- Prevents epidemic explosion.

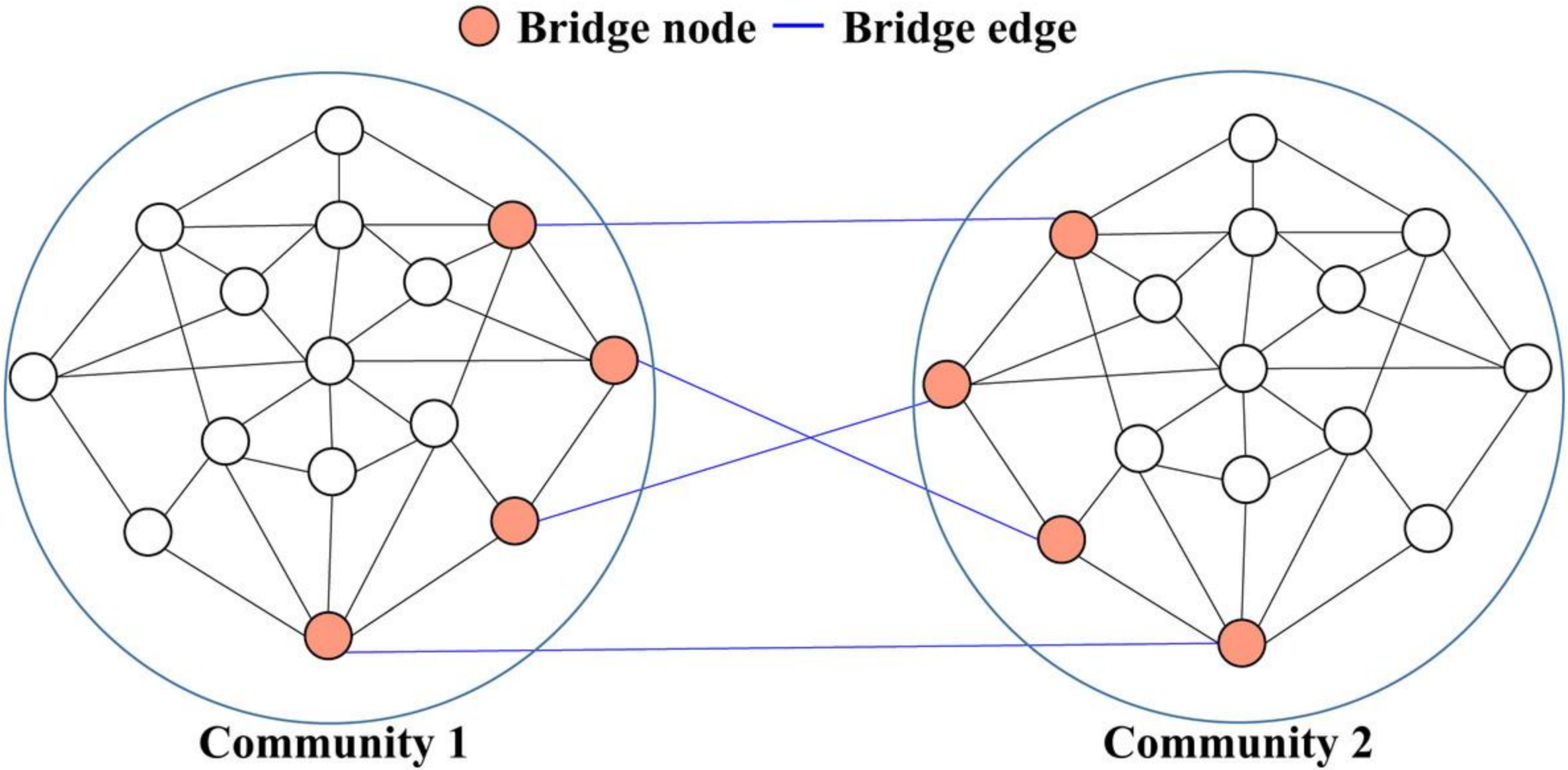
Immunize Hubs Strategy
Metric: Degree centrality
Definition: Protect or remove highest-degree nodes.
Why: Prevents rapid large-scale spread.

Summary table

Strategy	**Metric**	**Goal**
Seed hubs	Degree	maximize spread
Seed bridges	Betweenness	cross communities
Immunize hubs	Degree	stop spread

Method for stopping Diffusion

- Remove bridges → diffusion cannot spread globally
- **Cut bridges strategy** removes or monitors edges (or nodes) with high betweenness centrality to prevent diffusion between communities.
- Metric used:
  - Edge betweenness centrality
  - Node betweenness centrality


Bridge node
Bridge edge
Community 1
Community 2

Why cutting bridges works

- Diffusion requires paths.
- Bridge edges provide shortest paths between communities.
- Removing bridge eliminates cross-community paths.
- Thus diffusion cannot spread globally.
- Before bridge removal, Path exists.
- After bridge removal, Communities disconnected.

How?

- **Cut Bridges Strategy**
- Metric: Betweenness centrality (edges or nodes)
- Definition: Remove or monitor bridge edges connecting communities.
- Why: Prevents cross-community diffusion.
- **Algorithm (Edge Betweenness)**
- Step 1: Compute edge betweenness centrality
  Step 2: Identify highest betweenness edges
  Step 3: Remove or monitor those edges

Real-world Use Case: Quarantine Strategy - Lockdown

- Communities: Cities, regions, countries
- Bridge edges: Travel routes, Flights, Highways
- Cutting bridges: Travel restrictions, Border control, Quarantine zones
- Result: Disease contained locally.
- **Example: COVID Lockdowns**
- International flights = bridge edges
- Closing flights prevented global spread.
- Local outbreaks isolated.

Summary table

Strategy	**Metric**	**Goal**	**Use Case**
Seed hubs	Degree	maximize spread	viral marketing
Seed bridges	Betweenness	cross communities	break echo chambers
Immunize hubs	Degree	stop spread	epidemic control
Cut bridges	Betweenness	isolate communities	quarantine

Weak Ties v/s Seed Bridges

- Strong ties give trust. Weak ties give new information.
- Weak ties are edges representing infrequent, low-strength, or distant relationships that connect different social groups.
- Weak ties have low strength but high structural importance.
- E.g. Best friends → same information
  New acquaintance → new opportunities
- Seed bridges are nodes with high betweenness centrality selected as diffusion starting points to enable cross-community spread.
- Weak ties = property of edges
  Seed bridges = strategy using nodes

Weak Ties v/s Seed Bridges

Feature	**Weak Ties**	**Seed Bridges**
What it is	Edge property	Node selection strategy
Metric	Tie strength	Betweenness centrality
Introduced by	Granovetter (1973)	Network diffusion research
Purpose	Explain information flow	Optimize diffusion
Level	Edge-level concept	Node-level concept
Role in diffusion	Enable cross-community paths	Initiate cross-community spread

The Strength of Weak Ties

- Inside an echo chamber: everyone talks to the same people, everyone knows the same information, opinions reinforce
- These are **strong ties**: frequent, redundant, high trust, but low novelty
- Adding a **weak tie**:
- occasional contact
- different group
- low redundancy
- high novelty
- (different circles, different info, new exposure)
- ➞ suddenly information crosses clusters
- strong ties = reinforcement, weak ties = innovation (Weak ties act as **bridges**)

Summary

- Weak ties act as bridges between communities, introducing new information and breaking echo chambers.
- Strong ties build trust; Weak ties bring ideas.
- To spread fast, use hubs.
- To spread far, use bridges.
- To stop spread, protect hubs or cut bridges

More Diffusion

- Information diffusion based on influence is divided into three categories:
- individual influence,
- community influence, and
- influence maximization

Community Influence

- A community is a group of people with some common properties. In social networks, individuals will form various communities on the basis of interests.
- Definition 2 - A community is a subset of the network in which the users are densely connected and have similar attributes, e.g., they like to play badminton, or their research area is similar.
- Although the structure of social networks will change over time, communities remain relatively stable.

Takeaway

- Networks shape belief
- Structure beats content
- Position beats persuasion
- Trust beats information
- “If everyone were identical, position alone would determine influence.”
- “But in the real world, people are NOT identical.”
- E.g. A **controlled experiment in theory -** Physics says:
- “Assume frictionless surface” - But friction exists.
- Why ignore it? - Because we want to understand **gravity first**, separately.
- Then we add friction later.

Remarks

- In SNA:
- **Step 1 (model)**
- Assume: identical nodes, identical behavior, identical probability
- Study: only network topology
- This answers: “What does structure alone do?”
- **Step 2 (reality)**
- Then add: personality, authority, wealth, content quality
- Now: structure + attributes interact

Cont.

- **1. Node attributes (who you are)**
- authority
- charisma
- wealth
- expertise
- reputation
- **2. Network position (where you sit)**
- hubs
- bridges
- well-connected nodes
- **Both matter. Always.**
- **Influence depends on both who you are and where you sit in the network.**

A weak or incompetent leader in a powerful position can cause damage

- Influence/outcomes in society come from:
- Outcome $= f(attributes, ' \ position)$
- Both are necessary.
- **President / PM / General / Director**
- They have: extreme structural reach (huge network position) and huge broadcast power
- But if:
- poor judgment, incompetence, bad decisions
- Then:
- outcomes are harmful
- So here: attributes dominate. //outcomes depend strongly on attributes
- Position only amplifies whatever attributes exist.

Position gives you reach. Character determines the outcome.

- good leader + powerful position → great impact
- bad leader + powerful position → disastrous impact
- Same structure.
- Different attributes.
- Very different results.
- Structure determines how far influence travels; Attributes determine whether that influence is good or bad.

Conclusion

- Diffusion is not controlled by what spreads. It is controlled by where it starts.
- Network structure determines reach.
- Position determines power - how far influence travels.
- Attributes determine outcome. (Character determines whether that influence is good or bad)
- Strong ties create trust. Weak ties create opportunity.